Technical specification for low-voltage main switchboard MSB-2

Client: Ingleby Example Housing Ltd

Site: North Site, Example Way, Samplebury

Enquiry reference: ENQ-3192

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Prepared by: Sample Building Services Consultants Ltd

1 General

1.1 This specification covers the design, manufacture, testing and delivery of one low-voltage main switchboard, reference MSB-2, for a new apartment block.

1.2 Where this specification and the drawings disagree, the supplier shall ask for clarification before pricing.

2 Standards

2.1 The assembly shall comply with BS EN IEC 61439-1 and BS EN IEC 61439-2.

2.2 Circuit breakers shall comply with BS EN IEC 60947-2.

3 Electrical characteristics

3.1 Rated operational voltage: 400 V.

3.2 Rated frequency: 50 Hz.

3.3 System earthing: TN-C-S.

3.4 Prospective fault level: 36 kA for 1 s.

4 Construction

4.1 Form of separation: Form 4a to BS EN IEC 61439-2.

4.2 Degree of protection: IP54 with all doors closed.

4.3 The overall height shall not exceed 2200 mm including plinth.

4.4 Cable entry: bottom entry for all incoming and outgoing cables.

5 Incoming supply and busbars

5.1 Incomer: one air circuit breaker, 1250 A, 4 pole, withdrawable, with electronic trip unit.

5.2 Main busbars: rated 1250 A, TP&N, copper, fully insulated.

5.3 The incomer breaking capacity shall not be less than the prospective fault level.

6 Outgoing ways

6.1 The board shall have 8 outgoing ways as scheduled in Table 6.1.

6.2 MCB ways are cascade protected by the upstream device.

Table 6.1 Outgoing ways

Way	Description	Device	Rating (A)	Poles	Icu (kA)	Cable
Q1	Small power DB-L2	MCB	20	3	10	4C 4 mm2 SWA
Q2	Workshop DB	MCCB	250	4	36	4C 120 mm2 SWA
Q3	Sprinkler pump	MCCB	100	4	36	4C 35 mm2 SWA
Q4	Spare	MCB	63	3	10	4C 16 mm2 SWA
Q5	Small power DB-L1	Switch-disconnector	250	4	-	4C 120 mm2 SWA
Q6	Server room UPS	MCB	20	3	10	4C 4 mm2 SWA
Q7	Lighting DB-L2	MCCB	125	3	36	4C 50 mm2 SWA
Q8	External lighting	MCCB	125	3	36	4C 50 mm2 SWA

7 Metering

7.1 The incomer shall be fitted with three current transformers, ratio 1250/5 A, feeding a kWh energy meter.

8 Cable entry and glands

8.1 Removable undrilled gland plates shall be fitted to every cable chamber.

8.2 Brass glands for armoured cable shall be supplied for every outgoing cable.

9 Documentation

9.1 General arrangement drawings, single line diagrams and wiring diagrams shall be submitted for approval before manufacture.

9.2 Two printed copies and one electronic copy of the operation and maintenance manual shall be supplied at handover.

9.3 As-built drawings shall be issued within four weeks of completion.

10 Environmental conditions

10.1 The assembly shall be suitable for indoor installation at an ambient temperature between -5 and 35 degrees Celsius with a 24-hour average not exceeding 35 degrees.

10.2 Altitude of installation does not exceed 1000 m.

11 Spares and tools

11.1 One set of special tools, keys and racking handles shall be supplied.

11.2 A list of recommended spares for two years of operation shall be included with the tender.

12 Training

12.1 The supplier shall provide one half-day training session for site staff on operation and safe isolation.

13 Finish

13.1 The enclosure shall be finished in light grey, powder coated, with a minimum film thickness of 60 microns.

13.2 Plinths shall be finished in black.

14 Delivery and installation

14.1 The assembly shall be delivered in transport sections no wider than 1600 mm.

14.2 Offloading and positioning on site are by others unless stated in the tender.

14.3 The supplier shall attend site to reconnect transport sections and torque busbar joints.

15 General notes

15.1 All prices shall be firm for 90 days from the tender return date.

15.2 Any deviation from this specification shall be listed in the tender.